

Original research

Shape-sensing robotic-assisted bronchoscopy with integrated mobile cone-beam CT for small nodules: results from the prospective multicentre CONFIRM study

Bryan C Husta,¹ George Z Cheng,² Hitesh Batra,³ Janani S Reisenauer,⁴ William M Bartek,⁵ Or Kalchier-Dekel ,⁶ Aline Zouk,³ Nirali Patel,⁷ Mohit Chawla,⁸ George A Eapen,⁹ Carlos A Jimenez,⁹ Robert P Lee,^{8,10} Matthew J Bott,^{10,11} Scott Oh,¹² Roberto F Casal ⁹

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/thorax-2025-223272>).

For numbered affiliations see end of article.

Correspondence to
Dr Roberto F Casal;
rfcasal@mdanderson.org

Received 6 March 2025
Accepted 10 November 2025
Published Online First
7 December 2025

ABSTRACT

Background The use of 3D imaging with navigation bronchoscopy has increased with the introduction of mobile cone-beam CT (mCBCT). Shape-sensing robotic-assisted bronchoscopy (ssRAB) integrated with mCBCT has been introduced to correct for CT to body divergence (CT-BD) during the biopsy of peripheral pulmonary nodules (PPNs).

Methods Prospective observational multicentre study, enrolling patients with a PPN of ≤ 20 mm suspicious for malignancy, assessing tool-in-lesion (TIL), diagnostic yield, sensitivity for malignancy and incidence of pneumothorax or airway bleeding. Non-malignant index biopsies were followed for 12 months. Descriptive statistics are reported for all variables along with 95% CI for primary endpoints.

Results 155 consecutive patients were enrolled at six centres. Median nodule size was 14.0 mm (IQR 11.0–17.0), with 54.8% of nodules (85/155) in the upper lobes, 25.8% (40/155) with CT bronchus sign present and median distance to the nearest pleural surface of 8.2 mm (IQR 1.8–20.0). TIL was achieved in 154/155 (99.4%, 95% CI 96.5 to 100.0) procedures. A median of two (IQR 1–3) mCBCT spins were performed, with median dose area product of 25.7 Gy • cm² (IQR 11.0–46.7). Diagnostic yield per the American Thoracic Society/American College of Chest Physicians (ATC/ACCP) definition was 89.0% (n=138/155; 95% CI 83.0 to 93.5) and sensitivity for malignancy was 91.5% (95% CI 86.4 to 96.5). No pneumothorax (0%, 95% CI 0.0 to 2.4) and two Nashville Grade three bleeding events were reported (1.3%, 95% CI 0.2 to 4.6).

Interpretation The first prospective multicentre study of integrated ssRAB and mCBCT for small peripheral nodules resulted in diagnostic outcomes similar to those reported for transthoracic biopsy with a more favourable safety profile.

Trial registration number ClinicalTrials.gov ID: [NCT05562895](https://clinicaltrials.gov/ct2/show/study/NCT05562895).

INTRODUCTION

Lung cancer remains a significant global challenge, despite notable progress made in screening, diagnosis and treatment. Optimised outcomes for

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Current literature in robotic bronchoscopy and cone-beam CT guidance is mostly made of retrospective or small prospective studies, with inconsistent definitions of diagnostic yield.

WHAT THIS STUDY ADDS

⇒ Our prospective, multicentre study with the use of strict definition of diagnostic yield highlights that the integration of these two technologies results in superior diagnostic yield, sensitivity for malignancy and safety outcomes.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OF POLICY

⇒ In addition to providing mediastinal staging, bronchoscopy can also accurately and safely achieve diagnosis of peripheral tumours, challenging the more frequently used CT-guided approach and prompting the need for comparative trials.

patients with suspicious nodules require an efficient pathway from detection to treatment, particularly around comprehensive diagnosis—biomarker identification and mediastinal staging when indicated—with a low rate of complications and the fewest diagnostic encounters possible.

Electromagnetic navigation bronchoscopy (EMNB) emerged as an improvement to traditional bronchoscopy, followed by robotic-assisted EMNB (EMN-RAB); however, diagnostic yield and sensitivity for lesions in the periphery were still suboptimal.^{1–3} Challenges with these approaches can be attributed in part to difficulty reaching the lung periphery, as well as the phenomenon of CT-to-body divergence (CT-BD).^{1,2,4} Inherent to all navigation technologies, CT-BD is the differences in terms of lung volume and shape between preprocedural CT and the actual patient while under general anaesthesia, which can lead to difficulties in nodule localisation.⁵

While radial bronchial ultrasound (rEBUS) has been used as a common tool to facilitate nodule



© Author(s) (or their employer(s)) 2026. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: Husta BC, Cheng GZ, Batra H, *et al.* *Thorax* 2026;**81**:267–275.

localisation, atelectasis, which has been found to affect approximately 90% of patients undergoing bronchoscopy with general anaesthesia, can create false positive rEBUS signals.⁶ A significant gap between rEBUS visualisation and diagnostic yield has been previously reported, suggesting that rEBUS alone may not provide guidance to aid in a successful biopsy.^{7,8} Several techniques have emerged to prevent atelectasis and mitigate the impact of CT-BD, including ventilation protocols and the use of alternative patient positioning.^{9–11} Furthermore, adjunct imaging has progressed from 2D fluoroscopy and rEBUS to include advanced technologies that allow us to demonstrate tool in lesion (TIL) such as augmented fluoroscopy, tomosynthesis and fixed or mobile cone-beam CT (CBCT or mCBCT). The latter can help correct for CT-BD, providing nearly real-time nodule location via intraprocedural CT. However, when these advanced imaging technologies do not communicate with the navigational platform, the proceduralists are required to assess the spatial relationship of target and catheter tip and mentally assess adjustments to accurately deploy the biopsy tool.

The use of shape-sensing robotic-assisted bronchoscopy (ssRAB) has grown in recent years, leveraging the small catheter diameter for reach into the periphery of the lung while providing catheter stability during tool passage; however, the inherent issue of CT-BD remains. The integration of ssRAB (Ion Endoluminal System, Intuitive) and mCBCT (Cios 3D Spin Mobile, Siemens Healthineers) provides a single user interface with automatic data transfer of images and information between both systems, enabling virtual target updates based on an intraprocedural cone-beam CT. Currently, there is no multicentre data describing outcomes of ssRAB integrated with mCBCT. Herein, we report outcomes from the CONFIRM study—the first prospective multicentre evaluation of integrated ssRAB and mCBCT for sampling peripheral small pulmonary.

METHODS

Study design

CONFIRM is a prospective, multicentre, observational study enrolling consecutive patients planned for an ssRAB with integrated mCBCT biopsy. Eligibility criteria limited the study population to patients with a single solid or semi-solid lung nodule ≤ 20 mm in the largest diameter suspicious for malignancy.

All aspects of the research study were conducted in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. The study was approved by each site's Institutional Review Board (ClinicalTrials.gov: NCT05562895).

Procedure

A preprocedure non-contrast chest CT scan was uploaded into the ssRAB planning software (PlanPoint, Intuitive) to create a 3D segmented airway model with a navigation pathway and anatomy border. All cases were performed under general anaesthesia, with recommended ventilation settings following VESPA.⁹

After a manual airway survey, the ssRAB system was docked, registration performed and the catheter navigated to the target nodule. rEBUS was used at the investigator's discretion prior to mCBCT use or tool deployment. A biopsy needle (Flexision Needle, Intuitive) was recommended as the first tool used for sampling; needle deployment occurred prior to the first use of a mCBCT 3D spin if feasible. A breath hold at peak inspiration during imaging was recommended.¹² 3D spin was performed and the scan was transferred to the ssRAB system via the integration feature. Based on the review of catheter position relative to the target nodule in all planes, the target and catheter locations were

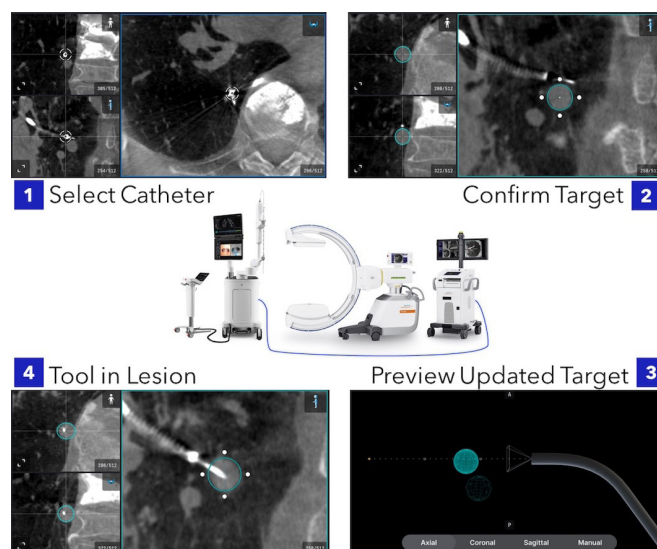


Figure 1 Integration of mobile cone beam CT and shape-sensing robotic bronchoscopy.

updated in the ssRAB system as needed. The integration process is described in figure 1. Sampling of the nodule was performed under 2D fluoroscopy using tools such as needles (Flexision, Intuitive), forceps, cytology brushes and/or a 1.1 mm or 1.7 mm cryoprobe (ERBECRYO2, Erbe) at the discretion of the physician. Rapid onsite evaluation was available at all sites. Confirmation of TIL with a 3D spin was mandatory per protocol. On completion of nodule sampling, a bronchoscopic airway exam was performed to assess for bleeding.

At the discretion of the operator, a mCBCT spin or fluoroscopy image was taken at the end of the procedure to assess for pneumothorax. Endobronchial ultrasound transbronchial needle aspiration (EBUS-TBNA) staging, if indicated, was performed after robotic bronchoscopy. A chest X-ray was recommended for all patients within 2 hours post-procedure. All patients were followed for approximately 30 days post-procedure for safety outcomes; any patient with a non-malignant index biopsy was followed for 12 months post-procedure.

Endpoints

The primary performance endpoints were TIL confirmation, diagnostic yield and sensitivity for malignancy. Primary safety endpoints included rates of pneumothorax and airway bleeding. Secondary endpoints included radiation dose and mCBCT procedure characteristics. Safety data were collected through 30 days post-procedure.

Definitions

Tool-in-lesion

TIL was defined as any portion of the needle visualised within the target in all three axes, assessed by the proceduralist during the procedure via mCBCT.

Diagnostic yield

We defined diagnostic yield as the proportion of patients with a malignant diagnosis or specific benign diagnosis, as listed in the American Thoracic Society/American College of Chest Physicians (ATS/ACCP) consensus, on the index biopsy only.¹³ The ATS/ACCP consensus statement was indeed published after the start of the study.¹³ Understanding that this is not a validated definition, we decided to still

adopt it since it is the most conservative one, and it will hopefully allow future researchers to use a common language.¹³ Our original diagnostic yield definition (selected prior to the advent of the ATS/ACCP consensus) was in line with the intermediate diagnostic yield definition and considered non-specific benign index biopsy findings diagnostic if they explained the presence of the nodule, fit the clinical presentation and were confirmed during radiographic follow-up.¹⁴ Both the ATS/ACCP and the 'intermediate' diagnostic yield definitions will be reported to allow for comparison with future studies as well as with previously published data. Biopsy results suspicious for malignancy, with atypia only, or with benign bronchial cells/lung parenchyma were considered non-diagnostic for both definitions.

Sensitivity for malignancy

We classified biopsy results as follows. Malignant samples on index biopsy were considered true positives. Non-malignant samples on index biopsy that were confirmed malignant within the 12-month follow-up with increased size, subsequent biopsy or resection or the start of empiric treatment were considered false negatives. Non-malignant samples on index biopsy that were confirmed benign within the 12-month follow-up with resolution, decreased size, stability ≥ 12 months or subsequent definitive sampling or excision were considered true negatives.

Safety

Safety outcomes assessed were pneumothorax with or without intervention, airway bleeding and any procedure-related adverse events. Airway bleeding was assessed according to the Nashville Bleeding Scale.¹⁵

Radiation dose

Radiation is measured using the dose area product (DAP), defined as the product of dose and beam area ($\text{Gy} \cdot \text{cm}^2$) and cumulative air kerma (CAK), defined as the total air kerma accumulated at the skin (mGy). Per recently published World Association for Bronchology and Interventional Pulmonology radiation reporting guidelines, doses for 2D and mCBCT are presented separately where available.¹⁶

Procedure times

Robotic procedure time was measured from the insertion of the ssRAB catheter into the endotracheal tube to the removal of the ssRAB catheter from the endotracheal tube. Navigation time was recorded from the time the registration was accepted until the time the virtual target was reached. Time to TIL was recorded from docking to first 3D spin demonstrating TIL.

Statistical analysis

Statistical analysis was performed using SAS V.9.4. This study was not powered for any hypothesis and ultimately a sample size was not statistically derived. This study describes the study population using measures of central tendency, spread and distribution for all studied variables, including demographics, nodule details, procedure characteristics and outcomes. Types of descriptive statistics were dependent on variable type as follows. Continuous variables were described with mean and SD, median and IQR. Discrete variables are described as counts and percentage of the study population. 95% Clopper-Pearson CIs are reported for primary endpoints. Sensitivity of the diagnostic yield primary endpoint to dichotomised subgroups of interest was assessed using the χ^2 test or, if any contingency table cell count was < 5 , Fisher's exact test. A p value of < 0.05 was considered statistically significant. Sensitivity analyses or imputation were deemed

Table 1 Demographics (n=155)

Variable	Results (n=155)
Age, years	
Mean ($\pm$ SD)	65.9 (12.8)
Median (IQR)	67.0 (58.0–75.0)
Range	20.0, 87.0
Sex, n (%)	
Female	87 (56.1)
Male	68 (43.9)
BMI, kg/m^2	
Mean ($\pm$ SD)	27.4 (6.0)
Median (IQR)	26.6 (23.4–30.4)
Range	16.2, 53.0
Comorbidities, n (%)	
≥ 1 comorbidity	120 (77.4)
Five most prevalent and clinically relevant *	
Hypertension	83 (69.2)
Chronic obstructive pulmonary disease	39 (32.5)
Coronary artery disease or history of myocardial infarction	27 (22.5)
Emphysema	24 (20.0)
Pneumonia within past year	15 (12.5)
Smoking status, n (%)	
Current	30 (19.4)
Previous	74 (47.7)
Never	51 (32.9)
ASA classification, n (%)	
I	1 (0.6)
II	22 (14.2)
III	123 (79.4)
IV	9 (5.8)
Preprocedure suspicion for malignancy, n (%)	142 (91.6)
Low	8 (5.6)
Intermediate	50 (35.2)
High	84 (59.2)
Preprocedure suspicion for benign process, n (%)	13 (8.4)
*The remaining clinically relevant comorbidities included (in descending order of prevalence): cerebrovascular disease, congestive heart failure, bronchitis/chronic bronchitis, pulmonary infection, pulmonary hypertension, bronchiectasis, pulmonary fibrosis, history of haemoptysis, interstitial lung disease, coagulopathy, other bleeding disorder.	
.ASA, American Society of Anesthesiologists; BMI, body mass index.	

unnecessary for the three variables with elevated missing data rates (DAP, distance from nearest blood vessel, airway generation count) as these do not affect the primary endpoints. Only one of the remaining study variables had a missing data rate of 1.9%, while the remaining variables' mean missing data rate was $< 1\%$.

RESULTS

A total of 155 patients were consecutively enrolled from February 2023 to December 2023 across six sites and 14 proceduralists (See online supplemental e- figure 1 for patient flow chart). Patient demographics and clinical characteristics are presented in [table 1](#).

Table 2 Nodule details

Parameter	Results (n=155)
Largest diameter, mm	
Mean ($\pm$ SD)	13.8 (3.6)
Median (IQR)	14.0 (11.0–17.0)
Range	5.0, 20.0
Lobe location, n (%)	
Left lower lobe	33 (21.3)
Left upper lobe	43 (27.7)
Right lower lobe	18 (11.6)
Right middle lobe	19 (12.3)
Right upper lobe	42 (27.1)
Distance from nearest pleural surface, mm	
Mean ($\pm$ SD)	12.1 (12.0)
Median (IQR)	8.2 (1.8–20.0)
Range	0.0, 50.0
Missing	1
Distance from chest wall, mm	
Mean ($\pm$ SD)	22.8 (22.1)
Median (IQR)	17.0 (5.0–34.3)
Range	0.0, 107.5
Missing	1
Distance from nearest critical blood vessel, mm	
Mean ($\pm$ SD)	14.6 (16.9)
Median (IQR)	7.5 (2.0–22.5)
Range	0.0, 75.0
Missing	31
CT bronchus sign, n (%)	
Absent	115 (74.2)
Present	40 (25.8)
Bronchus sign location, n (%)	
Airway adjacent to lesion	21 (52.5)
Airway leads directly to lesion	19 (47.5)
Density, n (%)	
Solid	132 (85.2)
Semisolid	17 (11.0)
Cavitary	6 (3.9)
Airway generation count, n	
Mean ($\pm$ SD)	7.1 (1.8)
Median (IQR)	7.0 (6.0–8.0)
Range	3.0, 13.0
Missing	14
Bullae present in the same lobe, n (%)	
No	145 (93.5)
Yes	10 (6.5)

Nodule details are presented in [table 2](#). Median axial, coronal and sagittal dimensions were 12.0 mm (IQR 10.0–16.0), 12.0 mm (IQR 9.0–14.3) and 12.0 mm (IQR 9.0–14.0), respectively; the overall median largest nodule diameter was 14.0 mm (IQR 11.0–17.0), with 85 out of 155 nodules in the upper lobes (54.8%). Median distance from the nearest pleural surface was

8.2 mm (IQR 1.8–20.0). CT bronchus sign was present in only 40 of 155 nodules (25.8%). Most nodules were solid (132 of 155, 85.2%) and located in the outer one-third of the lungs (88 of 155, 56.8%). Median airway generation count was 7 (IQR 6–8). Procedural details are summarised in [table 3](#).

Primary performance endpoint:

TIL was visualised in 152 cases (98.1%, 95% CI 94.5 to 99.6). Two additional cases, where the confirmatory spin was not performed but with a diagnostic biopsy, were considered TIL, leading to an overall TIL rate of 99.4% (95% CI 96.5 to 100.0). Median time from docking to TIL confirmation was 18.0 min (IQR 12.0–25.0).

Table 3 Procedural details

Variable	Results (n=155)
ssRAB procedure time, min*	
Mean ($\pm$ SD)	45.9 (16.4)
Median (IQR)	45.0 (35.0–56.5)
Range	13.0, 94.0
Missing	3
rEBUS attempted, n (%)	132 (85.2)
rEBUS signal, n (%)	
Concentric	44 (33.3)
Eccentric	68 (51.5)
No signal	20 (15.2)
Tools used, n (%)	
19G Flexision Needle	15 (9.7)
21G Flexision Needle	67 (43.2)
23G Flexision Needle	99 (63.9)
Cytology brush	49 (31.6)
Forceps	68 (43.9)
Bronchoalveolar lavage	46 (29.7)
Cryoprobe	88 (56.8)
Common tool combinations (not use order), n (%)	
Needle, cryobiopsy	31 (20.0)
Needle, forceps	27 (17.4)
Needle, brush, bronchoalveolar lavage, cryobiopsy	21 (13.5)
Needle only	15 (9.7)
Needle, forceps, cryobiopsy	14 (9.0)
Distance from catheter tip to near edge of nodule, mm	
Mean ($\pm$ SD)	5.7 (4.1)
Median (IQR)	5.0 (3.0–8.0)
Range	0.0, 18.0
Missing	1
Lymph node staging (n=107), n (%)	
No	32 (29.9)
Yes	75 (70.1)

*Procedure time was defined as the time the ssRAB catheter crossed the endotracheal tube (ETT) to the time the ssRAB catheter was removed from the ETT. rEBUS, radial bronchial ultrasound; ssRAB, shape-sensing robotic-assisted bronchoscopy.

Table 4 Diagnostic characteristics

	Results n=155
ATS/ACCP diagnostic yield, n (%) (95% CI)	138 (89.0) (83.0 to 93.5)
Malignant, n (%)	107 (69.0)
Primary lung cancer	79
Adenocarcinoma	37
Atypical carcinoid	2
NSCLC: not specified	4
Small cell carcinoma	2
Squamous cell carcinoma	17
Typical carcinoid	9
Other	8
Metastatic tumour	28
Breast	3
Colorectal	3
Head/neck	6
Kidney	3
Melanoma	4
Thyroid	1
Other origin	8
Benign, n	34
Specific benign, n (%)	31 (20.0)
Infectious	19
Bacterial	11
MAC infection	4
MAI infection	1
MAB infection	1
Staphylococcus	2
Streptococcus pneumoniae	2
Pneumonia	1
Fungal	8
Aspergillosis	5
Rhodotorula mucilaginosa*	1
Coccidiomycosis	1
Rare silver positive - fungal	1
Pulmonary hamartoma	4
Organising pneumonia	3
Granuloma	2
Granulomatous inflammation	1
Bronchial adenoma	1
Benign intrapulmonary lymph node *	1
Non-specific benign †, n (%)	3 (1.9)
Fibrosis	2
Acute inflammation and necrotic debris	1
Non-diagnostic, n (%)	14 (9.0)
Intermediate diagnostic yield, n (%) (95% CI)	141/155 (91.0)(85.3 to 95.0)
Sensitivity for malignancy, n (%) (95% CI)	107/117 (91.5)(86.4 to 96.5)
Negative predictive value, n (%) (95% CI)	37/47 (78.7)(67.0 to 90.4)
Prevalence of malignancy, n (%)	117/155 (75.5)

Continued

Table 4 Continued

	Results n=155
Clinical stage ‡ (n=67), n (%)	
IA1	15 (22.4)
IA2	36 (53.7)
IA3	4 (6)
IB	5 (7.5)
IIB	1 (1.5)
IIIA	3 (4.5)
IIIB	1 (1.5)
IVB	2 (3)
Adequacy for molecular testing (n=56), n (%)	55 (98.2)
*Combined clinical-histopathological diagnosis; <i>Rhodotorula mucilaginosa</i> : highly complex immunocompromised patient in which the proceduralist and an infectious disease specialist determined this result was diagnostic and treated the patient with antifungals. Benign intrapulmonary lymph node : the pathologist determined the histopathology suggested an intra-pulmonary lymph node, with which the proceduralist agreed.	
†Non-specific benign diagnoses were not considered diagnostic for the ATS/ACCP diagnostic yield calculation.	
‡AJCC/UICC 8 th Edition Lung Cancer Stage Classification ¹⁷	
AJCC/UICC, American Joint Committee on Cancer/Union for International Cancer Control; MAB, mycobacterium abscessus; MAC, <i>Mycobacterium avium</i> complex; MAI, mycobacterium avium-intracellulare; NSCLC, non-small cell lung cancer.	

Primary performance endpoint: diagnostic yield

Diagnostic yield per ATS/ACCP recommendations was 89.0% (95% CI 83.0 to 93.5).¹³ All index diagnoses are listed in table 4. 69% of index biopsies were malignant, and 20.0% were specific benign. Diagnostic yield was comparable between the sites (range: 78.0%–100.0%, $p=0.50$). Subgroup analysis (figure 2) yielded no statistically significant differences ($p=0.48$ to 1.0) and no clinically significant performance differences (range 0pp–3pp) by subgroup.

Concurrent EBUS-TBNA staging was performed in 70.1% (75/107) of cases. Clinical stage for primary lung cancer was predominantly IA (82%; 55/67).¹⁷ When molecular testing was attempted (n=56), the sample was adequate and sufficient in 98.2% (55/56) of cases.

Primary performance endpoint: sensitivity for malignancy

Sensitivity for malignancy is 91.5% (107/117, 95% CI 86.4 to 96.5) with cancer identified in 117 subjects. One subject with a non-diagnostic index biopsy and one subject with a specific benign index biopsy (granuloma) were lost to follow-up; the minimum sensitivity estimate is 89.9% (107/119) if these two subjects are assumed to be false negatives for cancer.

Primary safety endpoint: rates of pneumothorax and airway bleeding

Zero (0.0%, 95% CI 0.0 to 2.4) pneumothoraces were reported. Six airway bleeding events occurred (3.9%, 95% CI 1.4 to 8.2), two of which were Nashville Grade 3 (1.3%, 95% CI 0.2 to 4.6). Five of the six airway bleeding events were associated with cryobiopsy, and one was associated with forceps biopsy. One of the two grade 3 bleedings required balloon occlusion of the target segment for 1 min due to brisk bleeding after cryobiopsy, and the other one, after forceps biopsy, required interruption of robotic procedure and temporary main stem intubation. In both cases, bleeding subsided spontaneously, and there was no hypoxaemia or haemodynamic instability. See further details in table 5.

Diagnostic Yield Subgroups

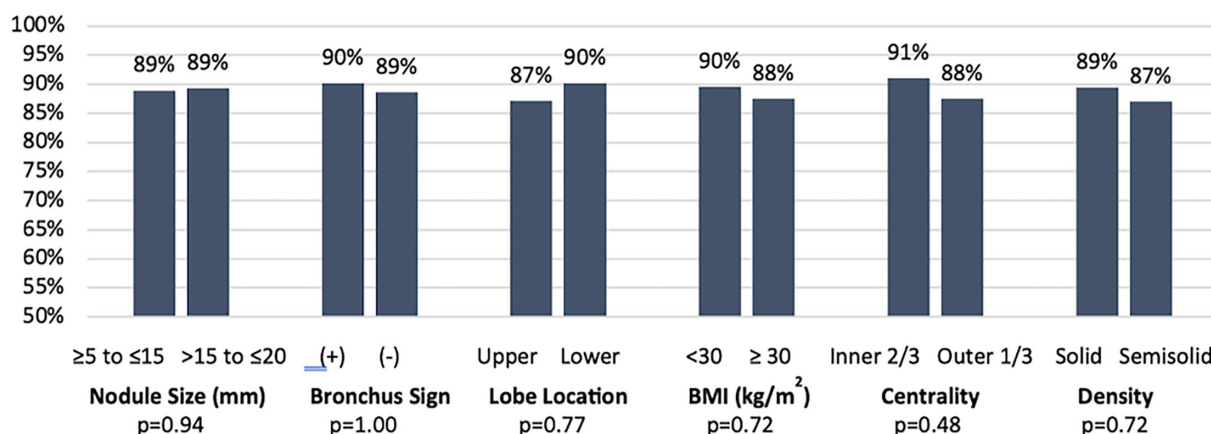


Figure 2 Diagnostic yield by subgroup. BMI, body mass index.

Secondary endpoints: radiation dose and CBCT characteristics

Median fluoroscopy time was 5.5 min (IQR 2.6–8.4), median CAK was 184.2 mGy (IQR 74.5–316.3), and the median DAP was 25.7 Gy•cm² (IQR 11.0–46.7). See further details in table 6 and online supplemental e-table 1.

DISCUSSION

The first prospective, multicentre study of ssRAB with integrated mCBCT demonstrates a very high diagnostic yield following ATS/ACCP definition, a high sensitivity for malignancy and nearly absolute TIL. The study also shows an excellent safety profile, with no pneumothorax, a low rate of serious bleeding events and low radiation exposure. Despite enrolling challenging nodules (small, mostly without bronchus sign, close to the pleura, away from the airways), subgroup analysis failed to find differences in diagnostic yields between subgroups by patient and nodule characteristics (eg, body mass index, bronchus sign presence, rEBUS signal), as well as between institutions and users, indicating reproducibility of outcomes. The integration of mCBCT to ssRAB allowed the proceduralist to correct for CT-BD and achieve strong clinical outcomes.

Table 5 Safety outcomes

Variable	Results (n=155)
AE type, n (%) (95% CI)	
Pneumothorax	0 (0.0) (0.0 to 2.4)
Airway bleeding	6 (3.9) (1.4 to 8.2)
Nashville grade 1	1 (0.6)
Nashville grade 2	3 (1.9)
Nashville grade 3	2 (1.3)
Other AEs *	6 (3.9) (1.4 to 8.2)
AEs related to the ssRAB system, n (%)	0 (0.0)
Premature termination, n (%) †	1 (0.6)
*Other AEs reported: prolonged (> 1 min) hypoxaemia, wheezing, shortness of breath, urinary retention, low blood pressure and soreness in chest. One patient experienced two adverse events for a total subject level AE count of 11.	
†One premature termination was reported associated with an intraprocedure grade 3 airway bleed; however, samples were obtained prior to the premature termination and the procedure was diagnostic.	
.AE, adverse event; SAE, serious adverse event; ssRAB, shape-sensing robotic-assisted bronchoscopy.	

Despite the relatively high pneumothorax rate and inability to perform nodal staging, CT-guided transthoracic biopsy remains a commonly performed biopsy approach, as well as the guideline-recommended technique, due to its perceived success rate.^{4 18–20} Factors increasing risk of pneumothorax, including emphysema, crossing bullae or a fissure and smaller and deeper lesions, may lead to patient selection bias for CT-guided biopsies and affect performance outcomes as a result.²¹ A meta-analysis from Balasubramanian *et al* reported an intermediate diagnostic yield for the CT-guided approach of 88.9% with a mean nodule size of 20.2 mm.²² Early data from the VERITAS study reported a median nodule size of 15.0 mm and comparable diagnostic yields between manual EMNB with tomosynthesis and CT-guided transthoracic biopsy groups of 76.0% when using the ATS/ACCP diagnostic yield definition, and a lower rate of pneumothorax requiring intervention in the EMN group (0.8% vs 8.8%).^{13 23} The CONFIRM results, in this context, suggest ssRAB+integrated mCBCT may be at least comparable for diagnostic performance with a favourable safety profile relative to the CT-guided approach, with the additional benefits of allowing biopsy of multiple nodules as well as concurrent EBUS-TBNA staging.

Data describing EMN-RAB performance outcomes are available in the literature. TARGET, the largest prospective study of EMN-RAB (17% with CBCT), reported 679 patients with a median nodule size of 20.9 mm, 59.9% bronchus sign presence, sensitivity for malignancy of 78.8% and a suboptimal ATS/ACCP diagnostic yield of 61.6%.²⁴ Cumbo-Nacheli *et al* published a series of 20 patients with EMN-RAB and CBCT, presenting a mean nodule size of 22 mm and a diagnostic yield of 63%, using a diagnostic yield definition similar to the ATS/ACCP consensus statement.²⁵ Shaller *et al* reported their experience with EMN-RAB+CBCT in 100 cases, with a 19 mm mean nodule size and an ATS/ACCP diagnostic yield of 91%.²⁶ This was, however, a retrospective report from a single centre with experienced proceduralists, and the authors classified as diagnostic six patients that would not be considered diagnostic per ATS/ACCP statement (three cases of fibroelastotic scar, one case of chronic haemorrhage with hemosiderosis and fibroelastotic changes and two cases of necrosis).

An ATS/ACCP diagnostic yield of 89.0% in this study, despite a TIL rate of 99.4%, highlights the existence of factors beyond navigation and localisation success which contribute to diagnostic yield, such as limitations of biopsy tools, pathologic

Table 6 Radiation and mCBCT characteristics

Variable	Results (n=155)
Time to TIL, min*	
Mean (±SD)	20.0 (11.6)
Median (IQR)	18.0 (12.0–25.0)
Range	2.0, 58.0
Missing	1
mCBCT spins to confirm TIL	
Mean (±SD)	1.7 (0.9)
Median (IQR)	2.0 (1.0–2.0)
Range	1.0, 5.0
Total mCBCT spins during procedure	
Mean (±SD)	2.1 (1.2)
Median (IQR)	2.0 (1.0–3.0)
Range	1.0, 6.0
Dose area product (DAP), Gy • cm ²	
Mean (±SD)	32.2 (27.7)
Median (IQR)	25.7 (11.0–46.7)
Range	0.5, 122.0
DAP from 2D fluoroscopy, Gy • cm ²	
Mean (±SD)	19.4 (20.1)
Median (IQR)	12.2 (5.4–25.7)
Range	0.1, 113.8
Missing	19
DAP from mCBCT, Gy • cm ²	
Mean (±SD)	11.6 (14.7)
Median (IQR)	6.4 (1.6–14.9)
Range	0.1, 79.9
Missing	19
DAP per mCBCT spin, Gy • cm ²	
Mean (±SD)	5.3 (4.4)
Median (IQR)	5.0 (1.4–8.0)
Range	0.1, 21.2
Missing	24
Fluoroscopy time, min	
Mean (±SD)	6.1 (3.6)
Median (IQR)	5.5 (2.6–8.4)
Range	1.1, 15.4
Cumulative air kerma, mGy	
Mean (±SD)	231.2 (209.9)
Median (IQR)	184.2 (74.5–316.3)
Range	3.8, 1047.0

*Time to tool-in-lesion was calculated from the time the ssRAB catheter crossed the ETT to the time tool-in-lesion was confirmed using mCBCT.

ETT, endotracheal tube; mCBCT, mobile cone-beam CT; ssRAB, shape-sensing robotic-assisted bronchoscopy; TIL, tool-in-lesion.

interpretation, intralesional heterogeneity, as well as prevalence of disease. Improvements in these areas may be able to close the gap between TIL and diagnostic yield. Nevertheless, TIL confirmation can increase proceduralist confidence in the obtained results, particularly for benign diagnoses. The sensitivity for malignancy in this study is 91.5%—preliminary data

from TARGET reported a sensitivity for malignancy for EMN-RAB of 81.8%, and a meta-analysis from Folch *et al* reported a pooled sensitivity for malignancy of 77% (95% CI 72 to 82) for 40 studies of EMN.^{24 27} Likely in part to the inclusion of several cancer centres in this study, the prevalence of malignancy was 75.5%, which is within the range of previously reported prevalence for malignancy in bronchoscopic literature of 55–88%.²⁸ The rate of concurrent EBUS-TBNA staging was high, presenting an optimised approach for diagnosis and staging of suspicious lung nodules.

CONFIRM is the first multicentre RAB study, showing zero pneumothoraces across all institutions. The combination of integrated advanced imaging, as well as the precise control and stability provided by the ssRAB platform, may contribute to a low pneumothorax rate, which is particularly notable given the challenging median nodule size (median 14 mm) and location (median distance to pleura 8.2 mm) in this study. These results are in line with those of Husta *et al* who described one pneumothorax in 475 (0.2%) patients in a large retrospective series of non-integrated ssRAB and mCBCT, and those of Bashour and colleagues who had no cases of pneumothorax in a prospective study with 67 patients and with the same non-integrated technologies.^{29 30} An airway bleeding rate (Nashville Grade ≥3) of 1.3% (n=2) was noted (and an overall bleeding rate of 3.9%), with one event occurring with cryobiopsy and one occurring with forceps biopsy. No significant bleeding was directly associated with needle-based sampling. For context, NAVIGATE, a large multinational study of EMN, reported a pneumothorax rate of 4.7% and an airway bleeding rate (Common Terminology Criteria for Adverse Events scale ≥ 2) of 1.7%, and in a meta-analysis of navigational bronchoscopy, Kops *et al* reported an overall bleeding rate of 2.1%.^{2 31}

The overall DAP of 25.7 Gy•cm² reported here is acceptable, with this dose being approximately equivalent to half of the radiation dose of a diagnostic chest CT (40–60 Gy•cm²).³² It is also at the lowest end of the spectrum of what has been previously reported with CBCT-guided navigational bronchoscopy.¹⁴ Interestingly, radiation dose was highly dependent on institution, with one institution with a median DAP a factor of 10 lower than the other five institutions (median 3.0 Gy•cm², IQR 2.3–3.9). This low-end outlier indicates the need for robust research into the optimisation of CBCT settings for patient and nodule characteristics guided by as low as reasonably achievable (ALARA) principles.

There are limited data describing integrated ssRAB and mCBCT, and the benefits of this integration are unknown. Deeply integrating these technologies may reduce the cognitive burden and time required for calculating the lesion to catheter relationship, improving catheter adjustments and reducing the number of spins required to obtain TIL. Two prospective studies have been published on the use of non-integrated ssRAB+mCBCT. Reisenauer *et al* published a pilot study of 30 nodules with a mean procedure time of 55.4 min and using mean 2.5 mCBCT spins.³³ Bashour *et al* published a prospective study of 67 patients, with a mean procedure time of 51 min and using median three mCBCT spins.²⁹ In comparison, in this study using ssRAB with integrated mCBCT, these same institutions that participated in CONFIRM had lower mean procedure times of 41 and 31 min respectively, with reduced median spins.

Limitations of this study include the number of academic institutions included (5/6), as well as the experienced nature of the proceduralists, limiting the generalisability of the study results. While competency in ss-RAB can be achieved after 25 procedures, data on the learning curve for combined ss-RAB

and mCBCT are not available.³⁴ Given the additional guidance provided by mCBCT and the ease of use of the software integration, we hypothesise that either a similar or even smaller number of procedures would be needed to achieve competency for physicians adopting these technologies. Another limitation of this study is that it was not designed with a powered hypothesis or comparative arm; therefore, these data could be considered hypothesis-generating for future comparative studies. Industry bias was mitigated through various mechanisms including protocol preregistration, the use of standardised endpoints, an open-data policy and the authors' training in ethical research. In addition, six of the 15 authors who had no conflicts of interest with the sponsor reviewed the data analysis and were involved in the manuscript preparation.

The CONFIRM study demonstrated the high diagnostic yield and sensitivity for malignancy for biopsies of small complex peripheral nodules using the integrated ssRAB and mCBCT systems, with no pneumothorax, short procedure time, high adequacy for molecular testing and low radiation exposure. The capabilities of ssRAB may be refined by the use of intraprocedure 3D imaging, approaching and potentially exceeding the historic performance and safety profile of the percutaneous CT-guided approach. Additional research, including comparative studies between integrated ssRAB with mCBCT and CT-guided percutaneous biopsy for small nodules, is warranted to corroborate this last assertion.

Author affiliations

¹Pulmonary, Allergy and Sleep Medicine, Mayo Clinic in Florida, Jacksonville, Florida, USA

²Medicine, Division of Pulmonary, Critical Care, and Sleep Medicine, University of California San Diego, La Jolla, California, USA

³Department of Critical Care Medicine, University of Alabama at Birmingham Health System Authority, Birmingham, Alabama, USA

⁴Thoracic Surgery, Mayo Clinic, Rochester, Minnesota, USA

⁵Pulmonary Medicine, St. David's South Austin Medical Center, Austin, Texas, USA

⁶Pulmonary and Critical Care, Memorial Sloan Kettering Cancer Center, New York, New York, USA

⁷University of California San Diego, La Jolla, California, USA

⁸Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, New York, USA

⁹Pulmonary Medicine, The University of Texas MD Anderson Cancer Center, Houston, Texas, USA

¹⁰Memorial Sloan Kettering Cancer Center, New York, New York, USA

¹¹Department of Surgery, Memorial Sloan Kettering Cancer Center, New York, New York, USA

¹²Section of Interventional Pulmonology, Keck School of Medicine at USC, Los Angeles, California, USA

Acknowledgements The authors thank all participating investigators and their staff for their involvement in this study. We would also like to acknowledge Intuitive for its sponsorship; specifically, we would like to thank Alison Gorski, MS, CCRP and Lindsay Spotanski, CCRP for their clinical trial operation support. We also thank Hui Wang, PhD and Meng Lu, MS (Lumbrita, LLC) for statistical analysis.

Contributors BCH, GZC, HB, JSR, WMB, SO and RFC were involved in conception and design. BCH, GZC, HB, JSR, WMB, OK-D, AZ, NP, MC, GAE, CAJ, RPL, SO and RFC performed acquisition, analysis and interpretation of data. BCH, GZC, HB, JSR, WMB, OK-D, AZ, NP, MC, GAE, CAJ, RPL, MJB, SO and RFC drafted and critically reviewed the manuscript. BH, GZC, HB, JSR, WMB, OK-D, AZ, NP, MC, GAE, CAJ, RPL, MJB, SO and RFC approved the final version of the manuscript and agreed to be accountable for all aspects of the manuscript. RFC is the ultimate guarantor.

Funding Intuitive Surgical provided study-specific research funding to the participating medical institutions and funding for independent statistical analysis.

Competing interests BCH: Research grants and consultant fees from Intuitive Surgical. GZC: Research grants from Intuitive Surgical; educational grants from Medtronic, Boston Scientific, Olympus, Cook, Pentax, Fujii, Microtech, Thoracent and Biodesix; consultant fees from Intuitive Surgical, Olympus, Cook, Pinnacle Biologics and Siemens. HB: Research grants and consultant fees from Intuitive Surgical. JSR: Research grants and consultant fees from Intuitive Surgical. WMB: Research grants and consultant fees from Intuitive Surgical. OK-D: Nothing to disclose. AZ: None

to disclose. NP: Consultant fees from Biodesix. MC: Consultant fees from Intuitive Surgical. GAE: None to disclose. CAJ: None to disclose. RL: None to disclose. MJB: Consultant fees from Merck, Obsidian, Intuitive and AZ. SO: Part-time employee at Intuitive Surgical. RFC: Research grants from Intuitive Surgical and Siemens. Consultant fees from Olympus, Intuitive Surgical, Siemens and Canon.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by Mayo Clinic Institutional Review Board, IRB Application # 22-007277MD Anderson Institutional Review Board, Study ID 2022-0715. Memorial Sloan Kettering Institutional Review Board/Privacy Board, Protocol # 23-063. St. David's Healthcare Institutional Review Board, IRB Study # 22-10-03UC San Diego Institutional Review Board, IRB # 805614WCG Institutional Review Board, Study # 1346514. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. Data are available on reasonable request. The data that support the findings of this study are available from the corresponding author (RFC), on reasonable request.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

ORCID iDs

Or Kalchiem-Dekel <https://orcid.org/0000-0001-9856-627X>

Roberto F Casal <https://orcid.org/0000-0002-6167-475X>

REFERENCES

- 1 Ost DE, Ernst A, Lei X, *et al.* Diagnostic Yield and Complications of Bronchoscopy for Peripheral Lung Lesions. Results of the AQUIRE Registry. *Am J Respir Crit Care Med* 2016;193:68–77.
- 2 Folch EE, Bowling MR, Pritchett MA, *et al.* NAVIGATE 24-Month Results: Electromagnetic Navigation Bronchoscopy for Pulmonary Lesions at 37 Centers in Europe and the United States. *J Thorac Oncol* 2022;17:519–31.
- 3 Zhang C, Xie F, Li R, *et al.* Robotic-assisted bronchoscopy for the diagnosis of peripheral pulmonary lesions: A systematic review and meta-analysis. *Thorax Cancer* 2024;15:505–12.
- 4 Ali MS, Ghori UK, Wayne MT, *et al.* Diagnostic Performance and Safety Profile of Robotic-assisted Bronchoscopy: A Systematic Review and Meta-Analysis. *Ann Am Thorac Soc* 2023;20:1801–12.
- 5 Pritchett MA, Bhadra K, Calcutt M, *et al.* Virtual or reality: divergence between preprocedural computed tomography scans and lung anatomy during guided bronchoscopy. *J Thorac Dis* 2020;12:1595–611.
- 6 Sagar A-ES, Sabath BF, Eapen GA, *et al.* Incidence and Location of Atelectasis Developed During Bronchoscopy Under General Anesthesia: The I-LOCATE Trial. *Chest* 2020;158:2658–66.
- 7 Tanner NT, Yarmus L, Chen A, *et al.* Standard Bronchoscopy With Fluoroscopy vs Thin Bronchoscopy and Radial Endobronchial Ultrasound for Biopsy of Pulmonary Lesions: A Multicenter, Prospective, Randomized Trial. *Chest* 2018;154:1035–43.
- 8 Chaddha U, Kovacs SP, Manley C, *et al.* Robot-assisted bronchoscopy for pulmonary lesion diagnosis: results from the initial multicenter experience. *BMC Pulm Med* 2019;19:243.
- 9 Salahuddin M, Sarkiss M, Sagar A-ES, *et al.* Ventilatory Strategy to Prevent Atelectasis During Bronchoscopy Under General Anesthesia: A Multicenter Randomized Controlled Trial (Ventilatory Strategy to Prevent Atelectasis -VESPATrial). *Chest* 2022;162:1393–401.
- 10 Kammer MN, Heideman BE, Maldonado F. Should We Start With Navigation or Endobronchial Ultrasound Bronchoscopy?: Insights From Monte Carlo Simulations. *Chest* 2022;162:265–8.
- 11 Lin J, Sabath BF, Sarkiss M, *et al.* Lateral Decubitus Positioning for Mobile CT-guided Robotic Bronchoscopy: A Novel Technique to Prevent Atelectasis. *J Bronchology Interv Pulmonol* 2022;29:220–3.
- 12 Pritchett MA, Lau K, Skibo S, *et al.* Anesthesia considerations to reduce motion and atelectasis during advanced guided bronchoscopy. *BMC Pulm Med* 2021;21:240.
- 13 Gonzalez AV, Silvestri GA, Korevaar DA, *et al.* Assessment of Advanced Diagnostic Bronchoscopy Outcomes for Peripheral Lung Lesions: A Delphi Consensus Definition of Diagnostic Yield and Recommendations for Patient-centered Study Designs. An Official American Thoracic Society/American College of Chest Physicians Research Statement. *Am J Respir Crit Care Med* 2024;209:634–46.

- 14 Vachani A, Maldonado F, Laxmanan B, *et al.* The Impact of Alternative Approaches to Diagnostic Yield Calculation in Studies of Bronchoscopy. *Chest* 2022;161:1426–8.
- 15 Folch EE, Mahajan AK, Oberg CL, *et al.* Standardized Definitions of Bleeding After Transbronchial Lung Biopsy: A Delphi Consensus Statement From the Nashville Working Group. *Chest* 2020;158:393–400.
- 16 Wijma IN, Casal RF, Cheng GZ, *et al.* Radiation Principles, Protection, and Reporting for Interventional Pulmonology: A World Association of Bronchology and Interventional Pulmonology White Paper. *Respiration* 2024;103:707–22.
- 17 Detterbeck FC, Boffa DJ, Kim AW, *et al.* The Eighth Edition Lung Cancer Stage Classification. *Chest* 2017;151:193–203.
- 18 Pyarali FF, Hakami-Majid N, Sabbahi W, *et al.* Robotic-assisted Navigation Bronchoscopy: A Meta-Analysis of Diagnostic Yield and Complications. *J Bronchology Interv Pulmonol* 2024;31:70–81.
- 19 DiBardino DM, Yarmus LB, Semaan RW. Transthoracic needle biopsy of the lung. *J Thorac Dis* 2015;7:S304–16.
- 20 Zhang W, Chen S, Dong X, *et al.* Meta-analysis of the diagnostic yield and safety of electromagnetic navigation bronchoscopy for lung nodules. *J Thorac Dis* 2015;7:799–809.
- 21 Huo YR, Chan MV, Habib A-R, *et al.* Pneumothorax rates in CT-Guided lung biopsies: a comprehensive systematic review and meta-analysis of risk factors. *Br J Radiol* 2020;93:20190866.
- 22 Balasubramanian P, Abia-Trujillo D, Barrios-Ruiz A, *et al.* Diagnostic yield and safety of diagnostic techniques for pulmonary lesions: systematic review, meta-analysis and network meta-analysis. *Eur Respir Rev* 2024;33:240046:173..
- 23 Lentz RJ, Frederick-Dyer K, Planz V, *et al.* Navigational bronchoscopy versus computed tomography-guided transthoracic needle biopsy for the diagnosis of indeterminate lung nodules: initial results from the veritas multicenter randomized trial. American Thoracic Society 2024 International Conference; May 2024 10.1164/ajrcm-conference.2024.209.1_MeetingAbstracts.A6665 Available: <https://www.atsjournals.org/doi/book/10.1164/ajrcm-conference.2024.C100>
- 24 Murgu S, Chen AC, Gilbert CR, *et al.* A Prospective, Multicenter Evaluation of Safety and Diagnostic Outcomes With Robotic-Assisted Bronchoscopy: Results of the Transbronchial Biopsy Assisted by Robot Guidance in the Evaluation of Tumors of the Lung (TARGET) Trial. *Chest* 2025;168:539–55.
- 25 Cumbo-Nacheli G, Velagapudi RK, Enter M, *et al.* Robotic-assisted Bronchoscopy and Cone-beam CT: A Retrospective Series. *J Bronchology Interv Pulmonol* 2022;29:303–6.
- 26 Shaller BD, Duong DK, Swenson KE, *et al.* Added Value of a Robotic-assisted Bronchoscopy Platform in Cone Beam Computed Tomography-guided Bronchoscopy for the Diagnosis of Pulmonary Parenchymal Lesions. *J Bronchology Interv Pulmonol* 2024;31:e0971.
- 27 Folch EE, Labarca G, Ospina-Delgado D, *et al.* Sensitivity and Safety of Electromagnetic Navigation Bronchoscopy for Lung Cancer Diagnosis: Systematic Review and Meta-analysis. *Chest* 2020;158:1753–69.
- 28 Ortiz-Jaimes G, Reisenauer J. Real-World Impact of Robotic-Assisted Bronchoscopy on the Staging and Diagnosis of Lung Cancer: The Shape of Current and Potential Opportunities. *Pragmat Obs Res* 2023;14:75–94.
- 29 Bashour SI, Khan A, Song J, *et al.* Improving Shape-Sensing Robotic-Assisted Bronchoscopy Outcomes with Mobile Cone-Beam Computed Tomography Guidance. *Diagnostics (Basel)* 2024;14:1955.
- 30 Husta BC, Menon A, Bergemann R, *et al.* The incremental contribution of mobile cone-beam computed tomography to the tool-lesion relationship during shape-sensing robotic-assisted bronchoscopy. *ERJ Open Res* 2024;10:00993-2023.
- 31 Kops SEP, Heus P, Korevaar DA, *et al.* Diagnostic yield and safety of navigation bronchoscopy: A systematic review and meta-analysis. *Lung Cancer (Auckl)* 2023;180:107196.
- 32 Nickoloff EL, Lu ZF, Dutta AK, *et al.* Radiation dose descriptors: BERT, COD, DAP, and other strange creatures. *Radiographics* 2008;28:1439–50.
- 33 Reisenauer J, Duke JD, Kern R, *et al.* Combining Shape-Sensing Robotic Bronchoscopy With Mobile Three-Dimensional Imaging to Verify Tool-in-Lesion and Overcome Divergence: A Pilot Study. *Mayo Clin Proc Innov Qual Outcomes* 2022;6:177–85.
- 34 Bott MJ, Toumbacaris N, Tan KS, *et al.* Characterizing a learning curve for robotic-assisted bronchoscopy: Analysis of skills acquisition in a high-volume academic center. *J Thorac Cardiovasc Surg* 2025;169:269–78.